SCIENTIST GUIDE

Ratio Viña 0.4.0 · Ratio Complexus

# The vineyard as a *complex adaptive system*: model, climate, equations and evidence

Technical documentation of Ratio Viña for research and teaching: the data model with the establishment stage, the viticultural climate indices, the weighted network and its statistics, the cladogram, sustainability, regeneration, risk, the outlook, validation against NetworkX and SciPy with a real vineyard, and how to reproduce every number.

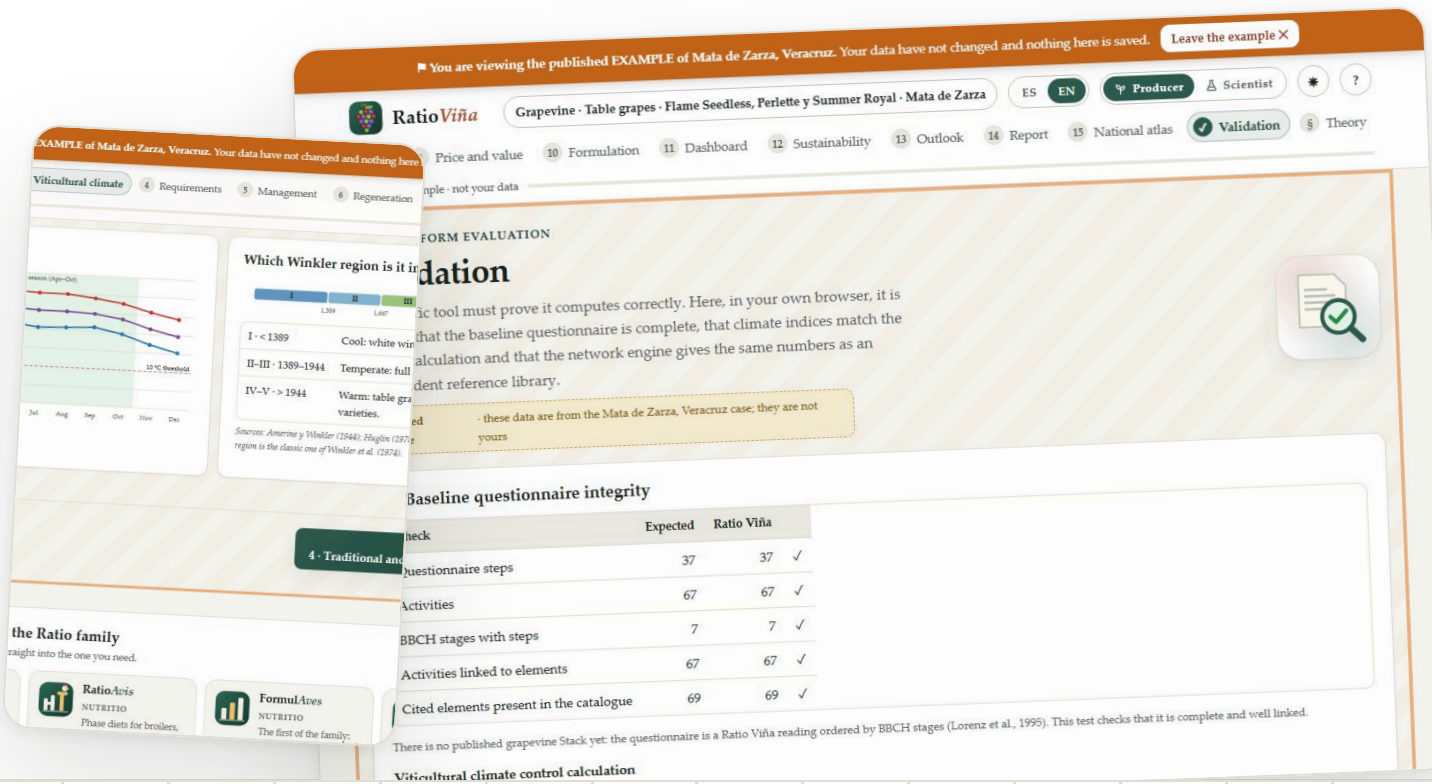
Open the app: **hectortecumshe-ai.github.io/RatioVina**

**1st edition** · October 2026 · covers Ratio Viña 0.4.0

DOI 10.5281/zenodo.24048803 · Text CC BY 4.0 · app MIT

Héctor Tecumshé Mojica-Zárate · Oscar Díaz-José · Edgar Arturo Sánchez-Moreno · Sergio Alejo-Bello · Luis Ángel Barrera-Guzmán

*For step-by-step use, see the Producer Guide.*




1 Vine
2 Region
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8 Elements
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FRONT MATTER

# Scope, data provenance and conventions

Ratio Viña represents a vineyard as a complex adaptive system (Holland, 1995): its elements are nodes and its management activities, the links. On that network it computes centralities, communities, correlations and a cladogram; characterises the viticultural climate with three published indices; assesses sustainability, regeneration and risk; and projects scenarios at 2, 5 and 10 years. Everything runs in the browser, in a single HTML file with no external dependencies.

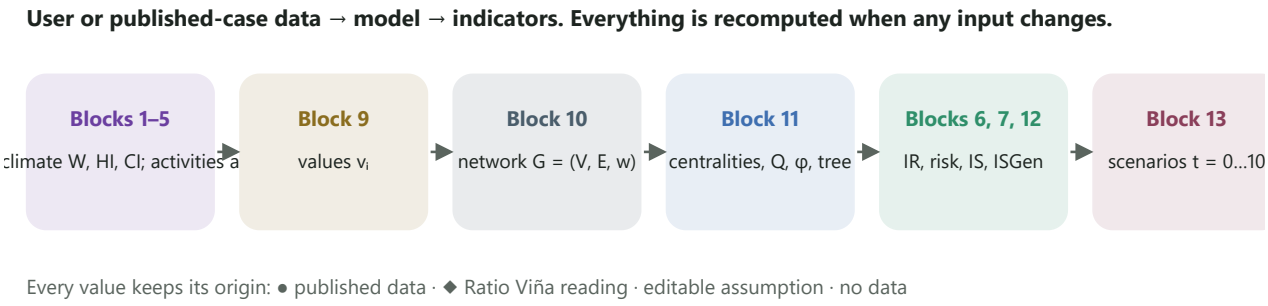



**FIGURE P.1** **The computation flow.** From activities to all other indicators.

# Provenance of every value

| LABEL | MEANING |
|---|---|
| **● Dato publicado** | Taken from a cited source: Mojica-Zárate et al. (2026), Lorenz et al. (1995), Amerine and Winkler (1944), Huglin (1978), Tonietto and Carbonneau (2004), INEGI (2020), Hersbach et al. (2020), IPCC (2006, 2018, 2019), Zhao et al. (2017). |
| **◆ Lectura de Ratio Viña** | Rule, grouping or value proposed by the app (for example, the annual-cycle questionnaire, activity–element links, impact profiles, class cut-offs). Editable. |
| **Supuesto** | Scenario parameter with no published basis (prices, costs, labour, risk reduction factors, grapevine β). Editable and flagged. |
| **Sin dato** | The value is not in the sources and was not entered by the user. The app does not impute it. |

| | | | |
|---|---|---|---|
| REFERENCE CASE | 1 ha cooperating plot, Mata de Zarza, Puente Nacional, Veracruz (30134-0018). Table grapes in establishment, January 2025 to February 2026 (Mojica-Zárate et al., 2026). | BASELINE QUESTIONNAIRE | 7 stages, 37 steps and 67 activities; 75 elements in 10 categories; 28 hazards. |
| TERRITORY AND CLIMATE | 32 states, 2,478 municipalities, 54,357 localities (INEGI, 2020); monthly temperatures from the user or, in the case, from ERA5 1991–2020. | IMPLEMENTATION | Library-free JavaScript; one file built from `src/` with `armar.py`. |

METHOD

# 1 Data model

Stages, activities, elements, value and impact: the inputs to every computation.

## 1.1 Stages and baseline questionnaire

There is no published local knowledge Stack yet for a producing vineyard in Mexico. The baseline questionnaire orders management into seven stages. The **establishment** stage (steps 1–6, 15 activities) was taken from the Mata de Zarza guide (Mojica-Zárate et al., 2026): choice of site and planting material, propagating cuttings, site preparation, planting, young-vine care and training. The other six (steps 7–37, 52 activities) are a Ratio Viña reading of the annual cycle ordered by the principal stages of the grapevine BBCH scale (Lorenz et al., 1995) with the operations in general viticulture texts (Winkler et al., 1974; Keller, 2020). Some activities are flagged as specific to a use (table or wine).

## 1.2 Activities and incidence

Each activity $a$ belongs to a step $p$ and a stage. It has a type (traditional or innovated) and an element set $E_a$. The incidence matrix $X$ (elements × activities) is 1 if the element takes part. Questionnaire activities are questions: only those answered "yes" and the innovated ones the user adds enter the computation; those answered "no" and unanswered ones stay out (nothing is preloaded). In the Mata de Zarza case 35 activities enter: 27 from the questionnaire and 8 innovated.

Steps are not fixed. Users can reorder them, change their stage, mark them "not applicable" (their activities leave $X$, the indices and the progress) and add their own steps. Each added activity carries its type (traditional or innovated), who proposes it (an actor or institution, which enters as an element of the activity) and since which year. Moving an activity to another step changes its step and stage, not its elements or impacts.

## 1.3 Element value

The user rates each element on four dimensions $k \in$ {economic, use, cultural, ecological}, from 0 to 5, with weights $\omega_k \in [0, 3]$. Without a rating the structural value, relative participation, is used:

$$v_i = \Sigma_k\, \omega_k \cdot c_{ik} / \Sigma_k\, 5\, \omega_k \quad \text{o bien} \quad v_i = n_i / \max_j n_j \tag{1}$$

where $c_{ik}$ is the rating and $n_i$ the number of activities element $i$ takes part in.

## 1.4 Impact by compartment

Five compartments $c$: soil, water, atmosphere, vegetation and fauna. Each element has an impact profile (a Ratio Viña reading with its rationale; for example, mulch +1 on soil and water); the activity's impact is the clipped sum, or the value edited by the user:

$$s_{ac} = \min(2, \max(-2, \Sigma_{i \in E_a} \pi_{ic})) \tag{2}$$

METHOD

# 2 Viticultural climate indices

## 2.1 Definitions

With monthly mean maximum $T_{x,m}$, minimum $T_{n,m}$ and mean $T_m$ temperatures (if missing, $T_m = (T_{x,m} + T_{n,m})/2$) and the days $d_m$ of each month:

$$W = \Sigma_{m\,=\,abr\ldots oct}\ \max(0,\ T_m - 10) \cdot d_m \tag{3}$$

$$HI = k\ \Sigma_{m\,=\,abr\ldots sep}\ \max(0,\ [(T_m - 10) + (T_{x,m} - 10)]/2) \cdot d_m \tag{4}$$

$$CI = T_{n,sep} \tag{5}$$

W: Winkler degree-days (Amerine and Winkler, 1944), with regions I (< 1389), II (1389–1667), III (1667–1944), IV (1944–2222) and V (> 2222). HI: Huglin's heliothermal index (1978), with $k$ = 1 below 40° latitude; Tonietto and Carbonneau (2004) classes: very cool ≤ 1500, cool 1500–1800, temperate 1800–2100, warm temperate 2100–2400, warm 2400–3000 and very warm > 3000. CI: Tonietto and Carbonneau's (2004) cool-night index for the northern hemisphere: very cool ≤ 12 °C, cool 12–14, temperate 14–18 and warm > 18. The app computes each index only if it has every month it needs.

! **VALIDITY OUTSIDE TEMPERATE CLIMATES**

The three indices assume a fixed growing season in the temperate northern hemisphere. If $\min_m T_m \geq 15$ °C, the app warns that the climate has no marked winter: in the tropics the vine may not go dormant, the cycle is driven by pruning and there may be two harvests a year (Kok, 2014). In such cases the indices serve to compare sites, not to zone varieties. This rule is a Ratio Viña reading.

## 2.2 Case data: ERA5 reanalysis

The Mata de Zarza guide publishes no temperatures, there is no National Weather Service station in the locality and the Huatusco Regional Academic Centre station belongs to another climate type. So the case uses the ERA5 reanalysis (Hersbach et al., 2020): daily temperatures from 1991 to 2020 in the nearest 0.25° cell (19.25° N, 96.75° W), adjusted to the locality's altitude (467 m) and downloaded from the Open-Meteo historical archive; monthly means average the 30 years. It is a model-based reconstruction, not an on-site measurement, and the project's "source" field says so.

| °C | E/J | F | M | A | M | J | J | A | S | O | N | D |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Max. | 23.7 | 25.4 | 27.5 | 29.8 | 30.7 | 30.1 | 29.8 | 29.8 | 29.0 | 27.9 | 25.9 | 24.3 |
| Min. | 14.7 | 15.5 | 17.1 | 19.3 | 20.9 | 21.5 | 20.5 | 20.8 | 21.2 | 19.7 | 17.2 | 15.3 |
| Mean | 19.1 | 20.3 | 22.2 | 24.5 | 25.7 | 25.6 | 25.1 | 25.1 | 24.8 | 23.7 | 21.4 | 19.1 |

With these values: W = 3,195 °C·d (region V), HI = 3,203 (very warm) and CI = 21.2 °C (warm nights); the coolest month averages 19.1 °C, so the no-marked-winter warning appears. The hand calculation is in chapter 12.

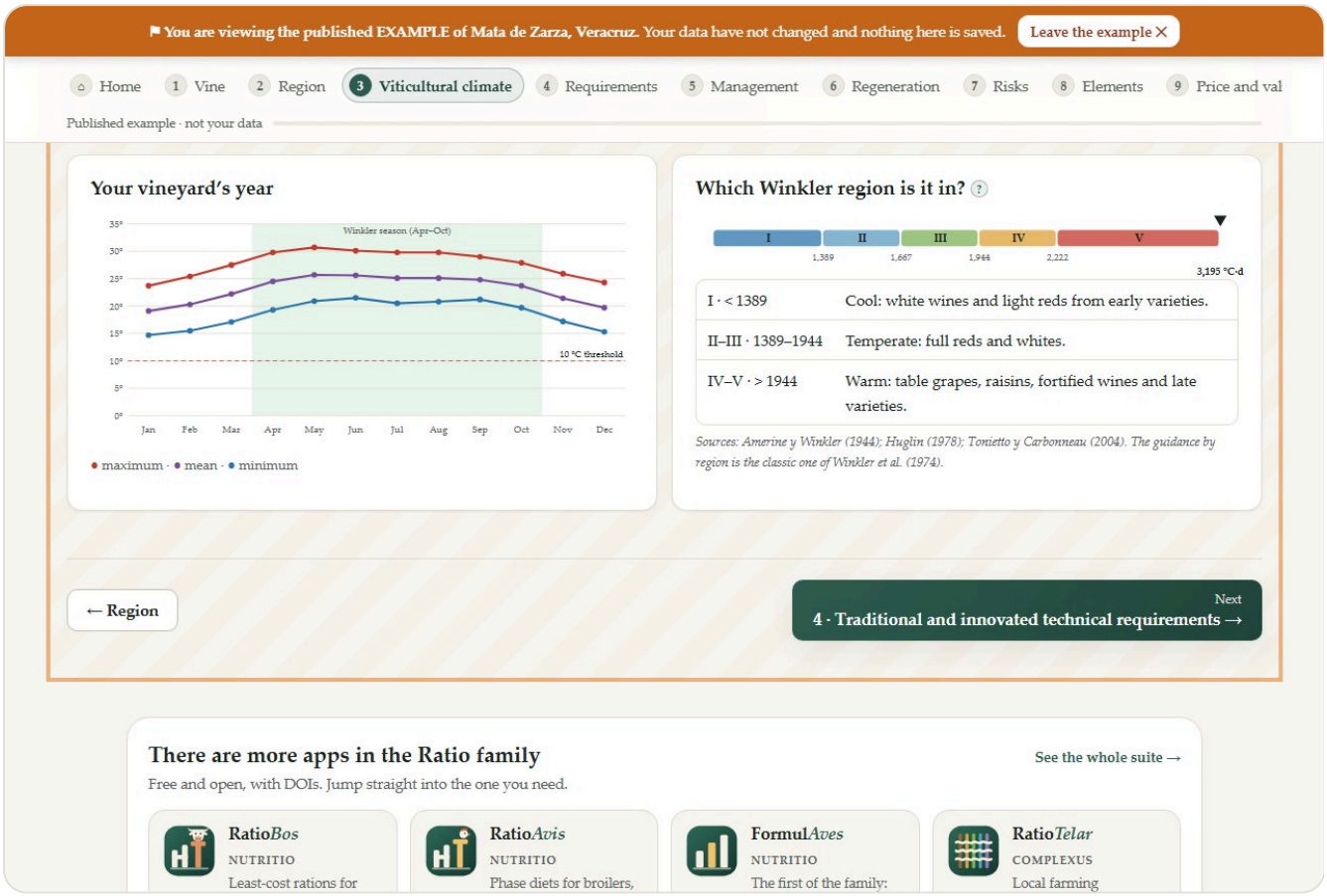



**FIGURE 2.1** **The Mata de Zarza vineyard's year** with ERA5 temperatures.

METHOD

# 3 The weighted network and its statistics

## 3.1 Network construction

Two elements are linked when they co-occur in an activity; the weight accumulates the geometric mean of their values over all shared activities, plus the catalogue's structural ties (for example, nurseries with cuttings, advisers with INIFAP), which enter only if they touch an element of the project's activities:

$$w_{ij} = \Sigma_{a\,:\,i,\,j \in E_a} \sqrt{(v_i \cdot v_j)} \qquad (6)$$

A zero value is replaced by 0.05 so the link is not cancelled. The "hide the omnipresent node" option excludes growers (PM), who take part in almost every activity.

## 3.2 Centralities

| MEASURE | DEFINITION IN THE APP |
|---|---|
| Weighted degree | $s_i = \Sigma_j\, w_{ij}$ |
| Betweenness | Brandes (2001) with Dijkstra and length $1/w_{ij}$; unnormalised (Freeman, 1978). |
| Closeness | Inverse of mean 1/w distance to reachable nodes, scaled by the reachable fraction (Wasserman & Faust, 1994). |
| Eigenvector | Power iteration on the weighted matrix (200 iterations, max normalisation). |
| Clustering | Unweighted local coefficient; mean path and diameter by BFS (Watts & Strogatz, 1998). |

## 3.3 Communities and modularity

Communities are detected with the Louvain method (Blondel et al., 2008), deterministic: local moves maximising ΔQ and aggregation, repeated until Q stops improving. Modularity (Newman, 2006):

$$Q = (1/2m)\ \Sigma_{ij}\ [\ w_{ij} - s_i\ s_j/2m\ ]\ \delta(c_i,\ c_j) \tag{7}$$

## 3.4 φ correlation

Pearson correlation between two elements' incidence vectors across the $N$ activities (equivalent to the φ coefficient of a 2 × 2 table):

$$\varphi_{ij} = \Sigma_a(x_{ia} - \bar{x}_i)(x_{ja} - \bar{x}_j)/(N\ \sigma_i\ \sigma_j) \tag{8}$$

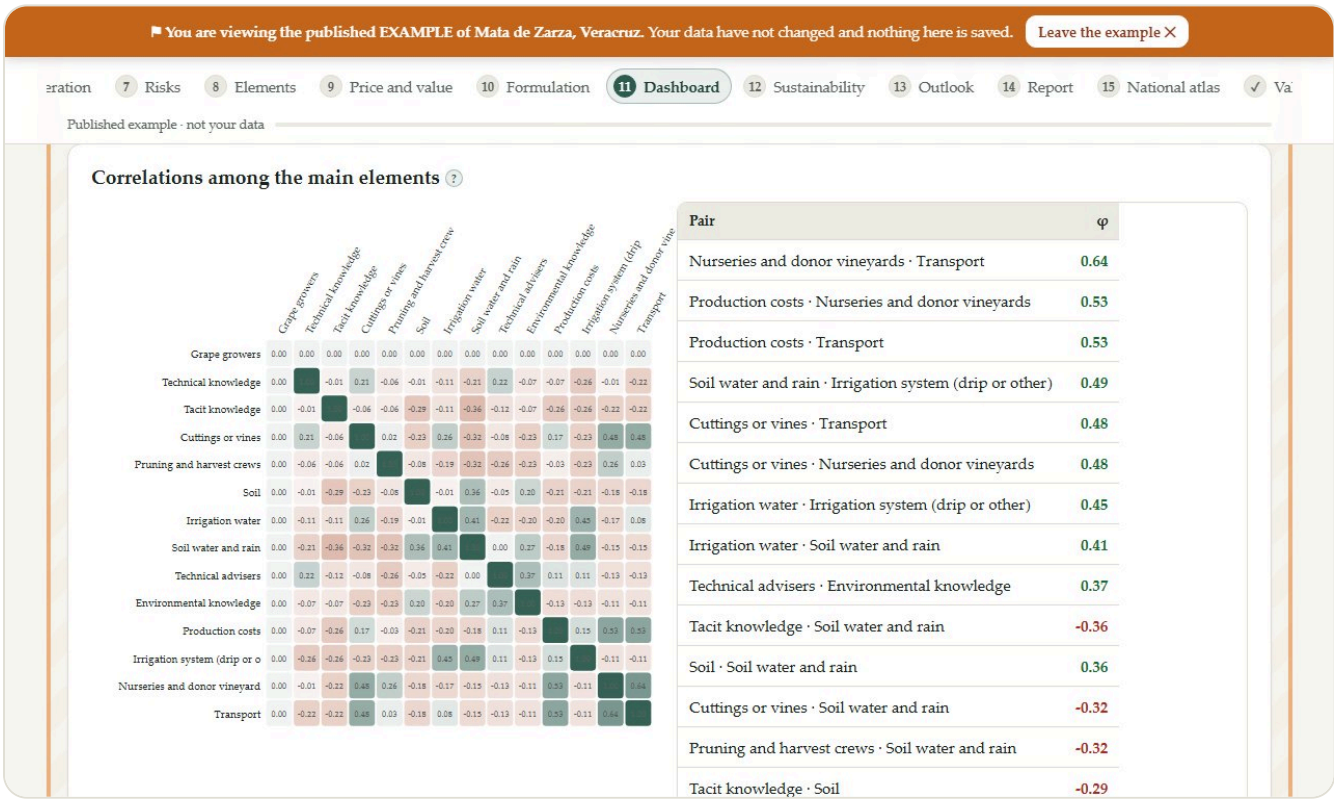



**FIGURE 3.1** **A Dashboard table in the scientist view.**

◆ **MATA DE ZARZA**

Full network: 42 nodes, 175 links, density 0.203, Q = 0.056. Without PM: 41 nodes, 136 links, Q = 0.1940 with 4 communities (soil, water and nutrition; knowledge and institutions; young-vine care and local knowledge; planting material), mean clustering 0.6761 and mean path 2.1329. The highest betweenness without PM belongs to technical knowledge (452.5), tacit knowledge (210.5), soil (202) and cuttings (160).

## 3.5 3D model and node importance

The 3D model in the Formulation block places each element on a layer according to its category: environment (disturbances), inputs (natural resources, inputs, equipment and financial resources), agents (actors, knowledge, institutions and policies) and outputs (products); management (5B) is a core without nodes. Within each layer, each category takes an angular sector proportional to its number of elements. Positions are for drawing only: links and their weights are those of the network (section 3.1), and the perspective is computed in the app, without libraries. The mapping of categories to layers is a Ratio Viña reading.

A node's importance is expressed from 0 to 100 relative to the network maximum, with the measure the user chooses:

$$I_i = 100 \cdot m_i / \max_j m_j \,,\; m \in \{ s, b, x, v \}$$

where $s$ is weighted degree, $b$ betweenness, $x$ the eigenvector (section 3.2) and $v$ the element value. The scale is relative to each network: it compares elements within a system, not across systems. By default the omnipresent node (PM) is hidden, because with weighted degree it dominates the scale: in Mata de Zarza, growers score 100 and the next element 31.

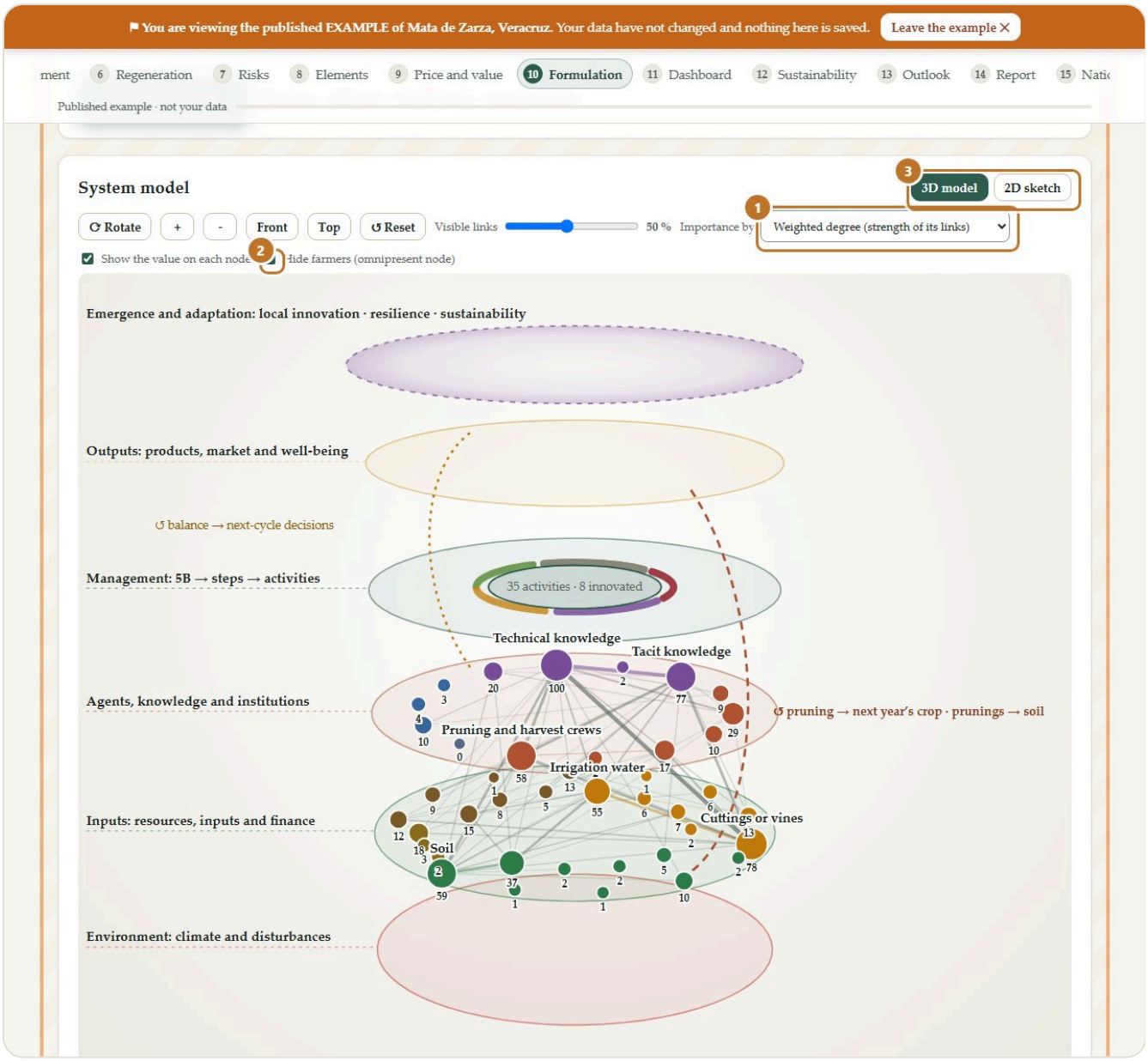



**FIGURE 3.2** **3D system model** in Mata de Zarza, without PM; each node's number is its weighted-degree importance.

### MATA DE ZARZA

Without PM, by weighted degree: technical knowledge (100), cuttings or vines (78), tacit knowledge (77), soil (59) and pruning and harvest crews (58). By betweenness: technical knowledge (100), tacit knowledge (47), soil (45) and cuttings or vines (35).

METHOD

# 4 Nearest-neighbour cladogram

## 4.1 Network distance

The distance between two elements is the shortest-path length in the full network, with length $1/w_{ij}$ per link (the same metric as betweenness and closeness). The path may pass through any element even if the tree shows a subset (actors; actors and institutions; all). Pairs with no path get 1.15 times the largest observed distance.

$$d_{ij} = \min_{\text{caminos } i \to j} \Sigma_{(u,v)} \, 1 / w_{uv} \tag{9}$$

## 4.2 Single linkage and cophenetic correlation

Agglomerative hierarchical clustering by single linkage (Sneath, 1957; Sokal & Sneath, 1963): at each step the two groups with the closest pair of members merge, $D(A, B) = \min_{i \in A,\, j \in B} d_{ij}$. Tree fidelity is measured with Sokal and Rohlf's (1962) cophenetic correlation, the Pearson correlation between $d_{ij}$ and the height at which $i$ and $j$ merge.

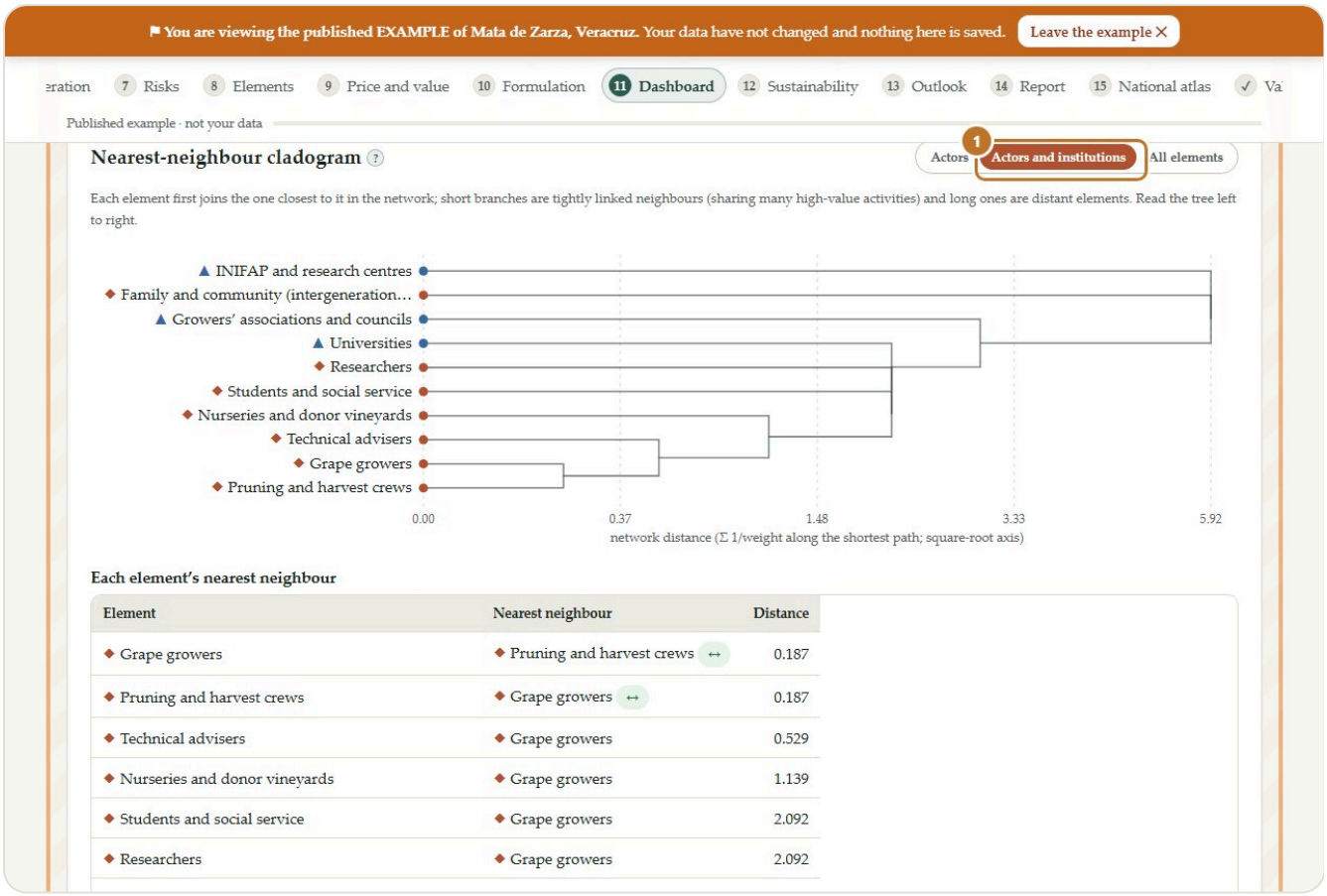



**FIGURE 4.1** **Mata de Zarza actors and institutions cladogram.** Growers and labourers are mutual neighbours.

| SUBSET | N | COPHENETIC R | \|ΔR\| VS SCIPY |
|---|---|---|---|
| Actors | 7 | 0.9646 | $3.3 \times 10^{-16}$ |
| Actors and institutions | 10 | 0.9199 | $1.1 \times 10^{-16}$ |
| All elements | 42 | 0.9750 | $7.0 \times 10^{-15}$ |

Merge heights match SciPy exactly in all three subsets (chapter 10).

> **?** **WHY SINGLE LINKAGE?**
>
> Because it directly answers "who is whose nearest neighbour?": each element's first merge is with its nearest neighbour, and the tree matches the minimum spanning tree of the distances.

METHOD

# 5 Sustainability, SDGs and carbon footprint

## 5.1 Activity sustainability index

$$IS_a = (1/5)\ \Sigma_c\ (s_{ac} + 2)/4 \quad \in [0, 1] \tag{10}$$

0.5 is neutral. The system IS is the mean of its activities; it is also reported by step and by type (traditional or innovated). In Mata de Zarza: 0.51.

## 5.2 Sarandón's general index

Following Sarandón (2002), each dimension is expressed on 0–4 through sub-indicators; their translation into app data is a Ratio Viña reading:

| DIMENSION | SUB-INDICATORS (0–4) | MATA DE ZARZA |
|---|---|---|
| IK | A margin 4(I − C)/I · B distinct products (cap 4) · C 4(1 − steps with external inputs/steps) | 1.65 (A sin dato, B 0, C 3.29) |
| IE | Mean of ($s_{ac}$ + 2) across activities and compartments | 2.03 |
| ISC | A stages with local knowledge · B stages with technical knowledge · C linked institutions | 2.93 (3.20, 3.20, 2.40) |

$$ISGen = (IK + IE + ISC)/3\ ; \quad \text{sostenible si } ISGen > 2 \text{ y } \min(IK, IE, ISC) \geq 2 \tag{11}$$

In Mata de Zarza, ISGen = 2.20, but IK = 1.65 < 2: not yet sustainable. IK-B is 0 because no practised activity produces grapes (the plot has not harvested) and IK-A stays as no data until income and costs are entered; IK averages the available sub-indicators.

## 5.3 SDG attainment

For each SDG the available evidence is averaged: environmental (mean of $(s_{ac} + 2)/4$ over the linked compartments: SDGs 2, 6, 12, 13 and 15) and structural (activities with technical knowledge or advice, SDG 4; linked institutions, SDG 17; margin, SDGs 1 and 8). Without evidence, "no data". The compartment–SDG mapping is a Ratio Viña reading. In Mata de Zarza 7 of the 17 have evidence (2, 4, 6, 12, 13, 15 and 17).

## 5.4 Carbon footprint per hectare

$$\mathrm{CO_2e} = 2.68\,D + N\,(0.010 + 0.11 \times 0.010 + 0.24 \times 0.011) \times {}^{44}\!/_{28} \times 273 + 0.20\,U \times {}^{44}\!/_{12} \tag{12}$$

$D$: litres of diesel; $N$: kg of N applied; $U$: kg of urea. IPCC default factors (2006; 2019 refinement): EF1, FracGASF × EF4 and FracLEACH × EF5; $N_2O$ GWP100 = 273 (AR6, Forster et al., 2021). Divided by yield it gives kg $CO_2e$ per tonne.

METHOD

# 6 Regeneration index

Eight ordinal indicators $s_k \in \{0, \ldots, 4\}$ group the four USDA-NRCS soil health principles (minimal disturbance, soil cover, diversity, living roots) and the processes the regenerative agriculture literature adds (livestock integration, fewer synthetic inputs, organic matter; Newton et al., 2020; Schreefel et al., 2020), plus local knowledge.

$$\mathrm{IR} = \Sigma_{k=1}^{8} s_k / 32 \quad \in [0, 1] \tag{13}$$

## 6.1 Suggested-value rules

| INDICATOR | RULE (RATIO VIÑA READING) | MATA DE ZARZA |
|---|---|---|
| Minimal tillage | n activities with plough, subsoiler or harrow (decisions excluded): 0→4, 1→3, 2→2, 3–4→1, ≥5→0 | 4 (n = 0) |
| Soil cover | 2 if there are activities with prunings, cover or mulch with positive soil impact; otherwise 1 | 2 |
| Diversity | species by system (pressurised irrigation 1, diversified 4, backyard 5…): 1→0, 2→1, 3→2, 4–5→3, ≥6→4 | 0 |
| Living roots | 3 since the vine is perennial | 3 |
| Livestock integration | 3 with animals; 2 if fodder is produced; otherwise 0 | 0 |
| Fewer synthetics | round(4 (1 − S⁄(S + B))) | 2 (S = 2, B = 3) |
| Organic matter | 1, 2 or 3 by residue-returning activities (0, 1, ≥2), +1 with soil testing | 2 |
| Local knowledge | min(4, round(8 f)), f = share of activities with tacit, cultural or family knowledge | 3 (f = 0.34) |

With the suggested values, IR = 16/32 = 0.50 (50, regenerative transition). Classes: < 0.25 degrading; 0.25–0.49 conventional; 0.50–0.74 regenerative transition; ≥ 0.75 regenerative. Trajectory is measured as ΔIR from a saved baseline and, with soil tests, as Δ% organic matter.

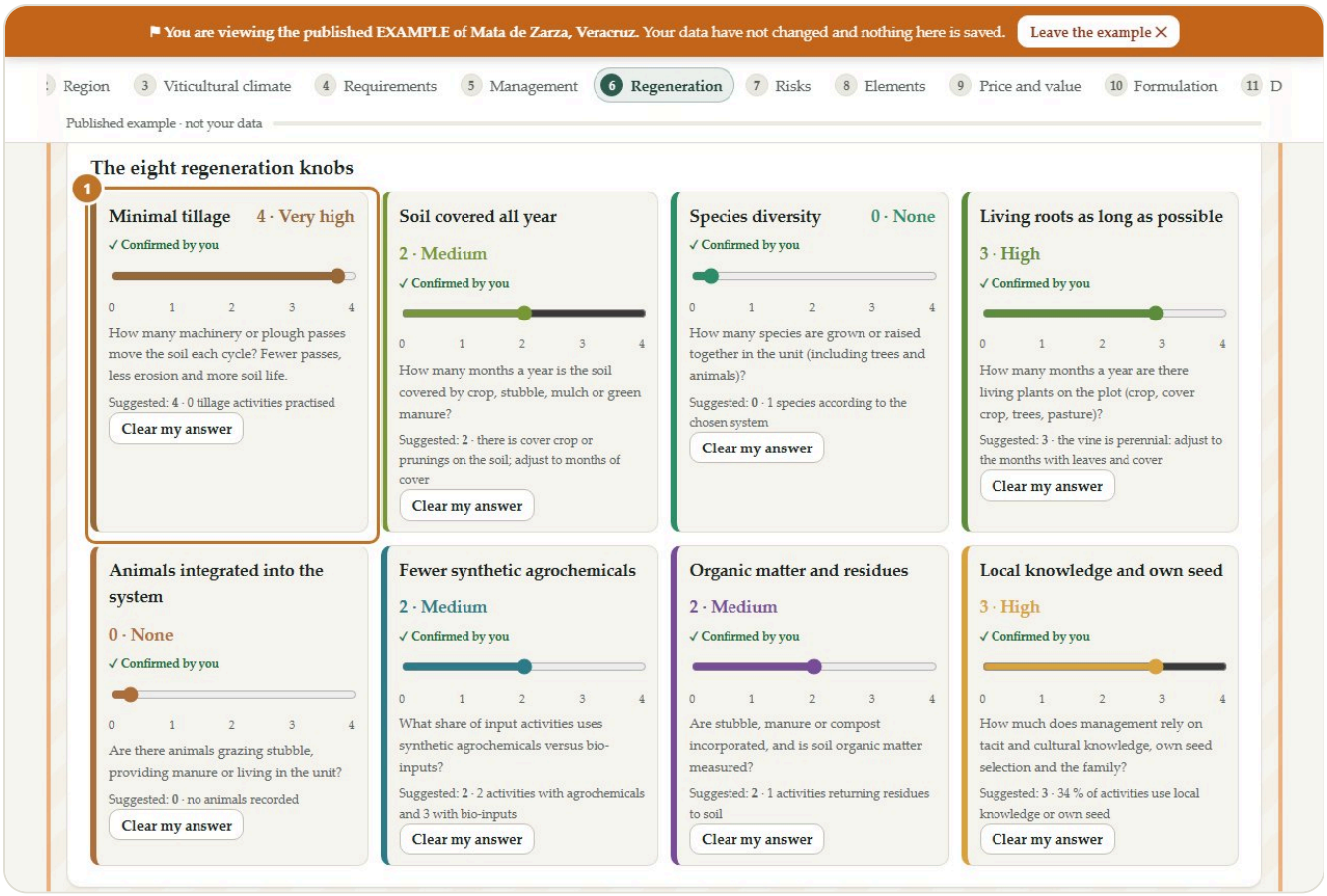



**FIGURE 6.1** **The eight indicators** with their suggested value and reason.

METHOD

# 7 Risk and adaptive capacity

Twenty-eight hazards $h$ in six categories. Each has likelihood $P_h$ and impact $I_h$ on 1–5 (5 × 5 matrix; IEC 31010:2019, within the ISO 31000:2018 process) and a control $c_h \in \{0, 1, 2, 3\}$. Following the IPCC (2014) framework, risk depends on the hazard and on vulnerability, which adaptive capacity reduces:

$$r_h = P_h\, I_h\, (1 - 0.4\ \mathrm{CA})(1 - 0.15\ c_h)\ ; \quad \text{índice} = \Sigma_h\, r_h / 25H \tag{14}$$

$$\mathrm{CA} = \mathrm{mean}(\mathrm{IR},\ \mathrm{IK_B}/4,\ \mathrm{ISC_B}/4,\ \mathrm{ISC_C}/4) \tag{15}$$

The 0.4 and 0.15 factors are Ratio Viña assumptions. Levels of $r$: ≤ 4 low, 5–9 medium, 10–15 high, ≥ 16 very high. Suggested values derive from the trend scenario (warming, costs, price, labour) and the share of activities with purchased inputs or local knowledge; the rest stays at 3.

## 7.1 Hazard catalogue

Each hazard is linked to the elements that transmit it (a Ratio Viña reading) to show risk in the Elements and Dashboard blocks. The nine added in version 0.2.0 show their source next to the name.

| CATEGORY | HAZARDS | SOURCE OF THE NEW ONES |
|---|---|---|
| Climatic (7) | Drought or late rains · frost · hail · heavy rain · heatwave · cyclones · **lack of winter chill** | Kok (2014) |
| Biological (8) | Pests · fungal diseases · **Pierce's disease** · **leaf-cutter ants** · **animal damage** · **diseased or poorly rooted material** · **herbicide drift** · weeds | Keller (2020); Mojica-Zárate et al. (2026); drift: Ratio Viña reading |
| Market (4) | Price fall · input cost rise · **new crop without a market** · middlemen | Mojica-Zárate et al. (2026) |
| Social (4) | Labour shortage and succession · **abandoning the crop** · loss of local knowledge · insecurity | Mojica-Zárate et al. (2026) |
| Institutional (2) | Withdrawal of programmes and support · tenure conflicts | — |
| Natural resources (3) | Erosion · water scarcity · **clay or poorly drained soils** | Mojica-Zárate et al. (2026) |

## 7.2 Suggested-value rules

| HAZARD | RULE |
|---|---|
| Drought, heatwave | $P = 4$ if trend warming is ≥ 0.2 °C per decade; otherwise 3 |
| Input cost rise | $P = 4$ if trend cost grows ≥ 2 %/yr; $I = 1 + \text{round}(4 \min(1, 2f_s))$, $f_s$ = share of activities with purchased inputs |
| Labour | $P = 4$ if $g_L \leq -1.5$ %; 3 if $g_L < 0$; otherwise 2 |
| Price fall | $P = 4$ if the trend price falls; otherwise 3 |
| Knowledge loss | $I = 1 + \text{round}(4 \min(1, 2f_k))$, $f_k$ = share of activities with tacit, cultural or family knowledge |
| All others | $P = 3$, $I = 3$ (moderate; adjust to the vineyard) |

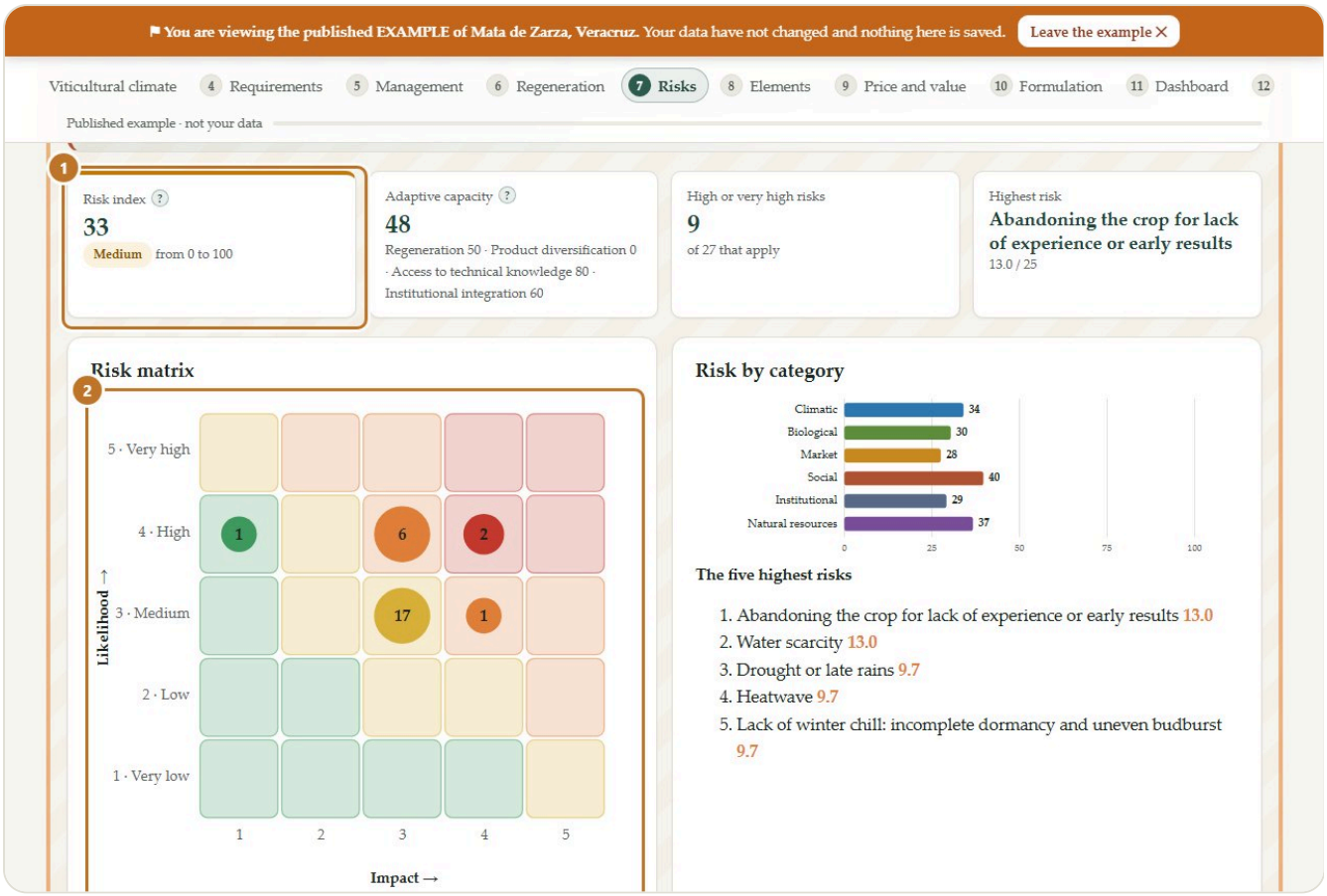



**FIGURE 7.1** **Mata de Zarza risk index and matrix.** Index 33, CA = 0.475.

◆ **MATA DE ZARZA**

CA = (0.50 + 0 + 0.80 + 0.60)/4 = 0.475. The hazards the guide describes carry values read from its text (◆): water scarcity and abandoning the crop, $P = 4$ and $I = 4$; leaf-cutter ants, new crop without a market and lack of winter chill, 4 and 3; animal damage, planting material and clay soils, 3 and 3. Frost is switched off. Hence 27 active hazards, 9 of high level and an index of 33.

METHOD

# 8 Scenario model

Three exploratory scenarios (Godet & Durance, 2011) are simulated year by year, $t = 0…10$. Adoption of innovated activities follows a logistic curve (Rogers, 2003) starting from the current share $a_0$:

$$A(t) = a_0 + (A_{máx} - a_0)\,[\sigma(t - t_{50}) - \sigma(-t_{50})] / [1 - \sigma(-t_{50})] \tag{16}$$

$$Y(t) = 100\,[1 + \beta\,\Delta T\,t/10] \cdot (1 - p(t)\,L_{ef}) / (1 - p_0\,L_{ef}) \cdot [1 + e_Y(A - a_0)] \cdot [1 - 0.5\,\text{déficit}(t)] \tag{17}$$

$$L_{ef} = L_{ext}(1 - 0.5\,IR_0)\;;\quad IR(t) = IR_0 + 0.5\,(1 - IR_0)(A - a_0) \tag{18}$$

β: yield change per °C. Zhao et al. (2017) publish it for maize, wheat, rice and soybean, not for grapevine; Ratio Viña uses −4.9 % as an editable assumption. ΔT: warming per decade (0.1, 0.2 and 0.3 °C; IPCC, 2018 likely range); $p(t)$: extreme-year probability, linear from $p_0$ to $p_{ext}$. Price and cost grow at constant rates; available labour declines at rate $g_L$ and required labour falls with adoption. Risk at $t$ recomputes (14) with $P_{climate} + 5(p(t) - p_0)$ and CA with $IR(t)$. The 0.5 coefficients are assumptions.

## 8.1 Price, cost, labour and sustainability

$$Pr(t) = 100\,(1 + g_P)^t\;;\quad C(t) = 100\,(1 + g_C)^t\,(1 - s_C\,\Delta A)\;;\quad \text{rel} = 100\,(Pr \cdot Y/100)/C \tag{19}$$

$$L_d(t) = (1 + g_L)^t\;;\quad L_n(t) = 1 - r_L\,\Delta A\;;\quad \text{déficit} = \max(0,\, L_n - L_d)/L_n \tag{20}$$

$$\mathrm{IS}(t) = \mathrm{IS}_{\mathrm{trad}}\,(1 - A) + \mathrm{IS}_{\mathrm{inn}}\,A \tag{21}$$

$\Delta A = A(t) - a_0$. If the user entered income $I_0$ and cost $C_0$ per hectare (Block 9), net income is $I_0\,(Pr \cdot Y/100)/100 - C_0\,C/100$.

## 8.2 Default parameters

| PARAMETER | DESIRABLE | TREND | UNDESIRED | BASIS |
|---|---|---|---|---|
| $g_P$ real price (%/yr) | +1 | 0 | −1.5 | assumption |
| $g_C$ real cost (%/yr) | +1 | +2 | +4 | assumption |
| ΔT (°C per decade) | 0.1 | 0.2 | 0.3 | IPCC (2018) |
| $p_{\mathrm{ext}}$ extreme year by year 10 | 0.12 | 0.18 | 0.25 | assumption |
| $g_L$ labour (%/yr) | −0.5 | −1.5 | −3 | assumption |
| $A_{\max}$ maximum adoption | 0.9 | 0.5 | 0.15 | assumption |
| $t_{50}$ half-adoption year | 4 | 6 | 9 | Rogers (2003) |

Common to the three scenarios: $p_0 = 0.10$; $L_{\mathrm{ext}} = 30$ % loss in an extreme year; adoption effect on yield $e_Y = 10$ %, on cost $s_C = 5$ % and on required labour $r_L = 15$ %. All are editable assumptions in the Block 13 assumptions table. Since $a_0 = 0.23 > A_{\max} = 0.15$ in the undesired scenario, adoption stays at $a_0$ there.

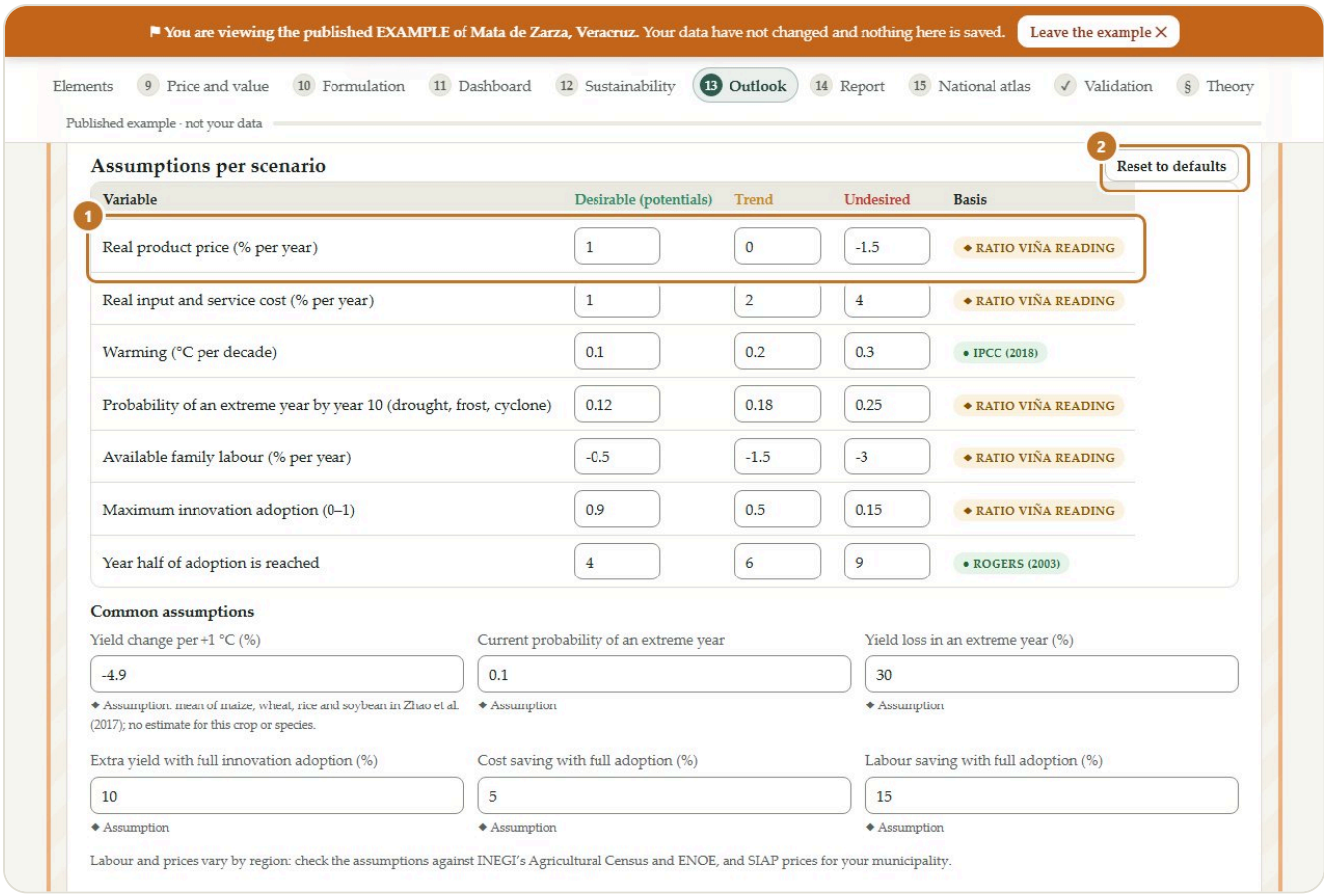



**FIGURE 8.1** **Assumptions table** with its basis: IPCC, Rogers or assumption.

| MATA DE ZARZA, T = 10 (TODAY = 100) | DESIRABLE | TREND | UNDESIRED |
|---|---|---|---|
| Yield | 106 | 95 | 83 |
| Gross income ÷ cost | 109 | 79 | 48 |
| Innovation adoption (%) | 90 | 50 | 23 |
| Regeneration index (0–100) | 67 | 57 | 50 |
| Risk index (0–100) | 32 | 33 | 34 |

METHOD

# 9 Territory and map

States, municipalities and localities come from INEGI's Geostatistical Catalogue and 2020 Population and Housing Census, with population, altitude and coordinates. Regions are Banco de México's (North, North-central, Central and South). Outlines are from Natural Earth (public domain). To place each locality on the map a projection was fitted:

$$x = 32.355\,(\lambda + 102)\,\cos^{0.4375}\varphi + 514.57\,; \quad y = -34.346\,\varphi + 1122.86 \tag{22}$$

95.4 % of localities fall inside their state outline; the point is approximate. Mata de Zarza: 19.230° N, 96.675° W, 467 m.

# 10 EVIDENCE AND USE
# Validation

## 10.1 Questionnaire integrity and climate control calculation

The Validation block checks that the questionnaire has 37 steps, 67 activities and 7 stages with steps, that every activity is linked to elements and that every cited element exists in the catalogue. For climate it uses a control case: with a constant mean of 20 °C from April to October, W = 10 × 214 days = 2140 °C·d (region IV); with a maximum of 28 °C from April to September, HI = [(20 − 10) + (28 − 10)]/2 × 183 days = 2562 (warm).

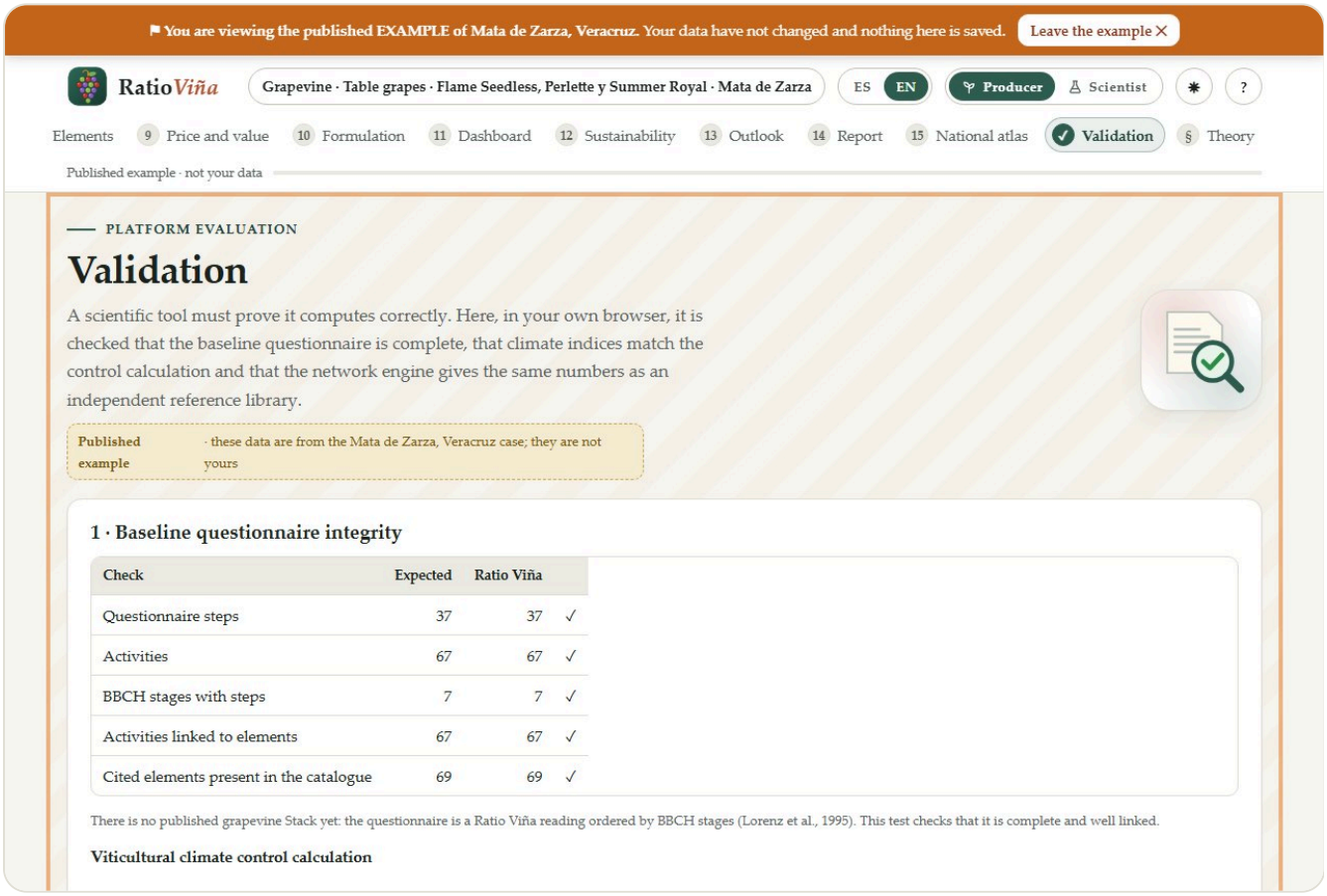



**FIGURE 10.1** **The app's validation block.**

## 10.2 Network engine and cladogram against NetworkX and SciPy

With Krackhardt's (1990) kite network, the app reproduces NetworkX betweenness, closeness, clustering, eigenvector and mean path with differences below 0.00005. In addition, the Mata de Zarza case network

was exported from example mode (`pruebas/red_matazarza.json`) and recomputed with NetworkX 3.7 and SciPy 1.18 (`pruebas/verificar_matazarza.py`):

| TEST | RATIO VIÑA | REFERENCE |
|---|---|---|
| Weighted betweenness (41 nodes without PM), max difference | | 0 |
| Q of the app partition (4 communities) | 0.1940 | 0.1940 |
| Mean clustering · mean path | 0.6761 · 2.1329 | 0.6761 · 2.1329 |
| Cladogram merge heights (3 subsets), max difference | | 0 |
| Cophenetic correlation (actors; actors and institutions; all) | 0.9646 · 0.9199 · 0.9750 | 0.9646 · 0.9199 · 0.9750 |
| Louvain Q (randomised NetworkX, seed 1) | 0.1940 | 0.2024 |

Metrics match exactly. NetworkX's randomised Louvain finds a four-community partition with slightly higher Q here (0.2024 versus 0.1940): Louvain is a heuristic and neither partition is necessarily optimal, so the reading of communities should be taken as exploratory.

EVIDENCE AND USE

# 11 Reproducibility and export

## 11.1 Code and tests

```bash
git clone https://github.com/hectortecumshe-ai/RatioVina
cd RatioVina
python armar.py                          # arma index.html desde src/
python pruebas/verificar_matazarza.py    # red y cladograma frente a NetworkX y
SciPy
```

## 11.2 Data for analysis

Block 14 exports to CSV the activities (with their five impacts and IS), nodes (value, participation, neighbours, weighted degree, betweenness, closeness, eigenvector, clustering, community), links (weight and number of activities), calendar, SDGs and cladogram (merges and nearest neighbours). **Save project** produces a JSON with the full state, which reproduces any result.

## 11.3 Deep links

The address accepts `#block/view/language`, for example …`/RatioVina/#clima/cien/en`. This guide's screenshots were taken this way, automatically.

## 11.4 Published case, migration and user data

The app starts blank. The Mata de Zarza case is loaded only in an **example mode** that is never saved (`estadoMataZarza()` in `src/18b_matazarza.js`: table grapes, own-rooted, rooted cutting planted on day 54 of 2025, 1 ha, 2,381 vines, ERA5 temperatures, 48 answered and 8 innovated activities, each with its source note); with it the network reproduces this guide's values. Structural ties enter only if they touch an element of the activities, in example mode too. Suggested regeneration and risk values count in the index, flagged as provisional, until the user confirms them. Project files carry `formato: 3`; version 0.1.0 files (`formato: 2`) are migrated on opening by shifting their steps from 1–31 to 7–37.

## 11.5 Scope and limits

- The reference case documents the establishment (first year) of a warm-climate vineyard; the flowering-to-post-harvest stages await a published Stack from a producing vineyard.

- The case temperatures are reanalysis (ERA5), not station data; in climates without winter the Winkler and Huglin indices are read with caution.
- Activity–element links, impact profiles, the SDG mapping, regeneration rules, the case's risk values and risk factors are Ratio Viña readings, flagged and editable.
- Scenarios are not forecasts; their value lies in comparing explicit assumptions. Grapevine β is an assumption.

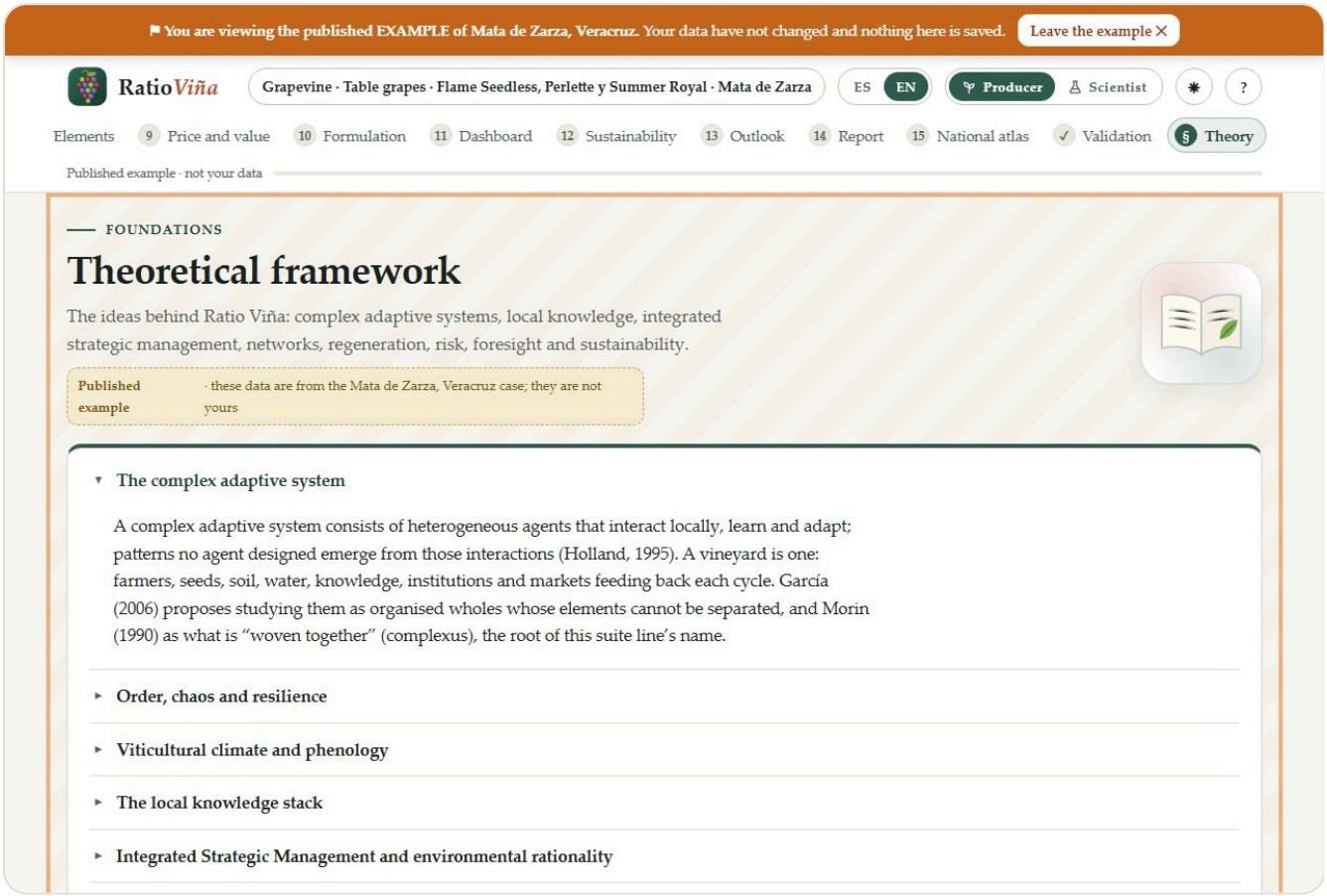



**FIGURE 11.1** **Theoretical framework** inside the app, with its bibliography.

EVIDENCE AND USE

# 12 Worked example: Mata de Zarza by hand

Every number in this chapter can be redone with a calculator and matches what app 0.4.0 shows.

## 12.1 Winkler degree-days

$$W = 14.5 \cdot 30 + 15.7 \cdot 31 + 15.6 \cdot 30 + 15.1 \cdot 31 + 15.1 \cdot 31 + 14.8 \cdot 30 + 13.7 \cdot 31 = 435 + 486.7 + 468 + 468.1 + 468.1 + 444 + 424.7 = 3194.6 \tag{3}$$

Region V (> 2222). The app shows 3195.

## 12.2 Huglin index

$$HI = 17.15 \cdot 30 + 18.2 \cdot 31 + 17.85 \cdot 30 + 17.45 \cdot 31 + 17.45 \cdot 31 + 16.9 \cdot 30 = 514.5 + 564.2 + 535.5 + 541.0 + 541.0 + 507.0 = 3203.1 \tag{4}$$

For example, April: $[(24.5 - 10) + (29.8 - 10)]/2 = 17.15$. Very warm class (> 3000). CI = $T_{n,sep}$ = 21.2 °C: warm nights.

## 12.3 Regeneration

$$IR = (4 + 2 + 0 + 3 + 0 + 2 + 2 + 3)/32 = 16/32 = 0.50 \tag{13}$$

Class: regenerative transition. If the plot added a cover crop with backyard poultry and the grower reflected it in the knobs (diversity 1 and animals 3), IR would reach 20/32 = 0.63.

## 12.4 Adaptive capacity and water-scarcity risk

$$CA = (0.50 + \%_4 + 3.^{20}\%_4 + 2.^{40}\%_4)/4 = (0.50 + 0 + 0.80 + 0.60)/4 = 0.475 \quad (15)$$

$$r_{agua} = 4 \times 4 \times (1 - 0.4 \times 0.475) \times (1 - 0.15 \times 0) = 16 \times 0.81 = 12.96 \quad (14)$$

High level (10–15); the app shows 13.0. Medium control ($c = 2$), for example 5,000 L ferrocement rainwater tanks as the guide proposes, would lower it to 12.96 × 0.70 = 9.07: medium level.

## 12.5 Sarandón general index

$$IK = (0 + 3.29)/2 = 1.65\ ; \quad ISGen = (1.65 + 2.03 + 2.93)/3 = 2.20 \quad > 2$$
$$\text{pero min} = 1.65 < 2 \ \Rightarrow\ \text{no sostenible aún} \quad (11)$$

The economic dimension is the main lever. Once the plot harvests, the harvest and sales activities add the table-grape product (IK-B = 1) and, with income and costs entered, margin IK-A enters the average.

## 12.6 Carbon footprint (illustrative example)

The guide publishes 40 g of diammonium phosphate (18 % N) per vine; with 2,381 vines that is 95.2 kg of diammonium phosphate and 17.1 kg of N per hectare. Without diesel or urea:

$$17.1 \times 0.01374 \times {}^{44}\!/_{28} \times 273 = 100.8 \text{ kg } CO_2e/\text{ha} \quad (12)$$

0.01374 = 0.010 + 0.11 × 0.010 + 0.24 × 0.011. It is an illustrative calculation with a single application; the case's real footprint needs the number of applications and the brush cutter's fuel, which the guide does not publish.

METHOD

# 13 National atlas aggregation

## 13.1 Summary of each vineyard

From each `.vina.json` file the atlas computes, with the app's own functions, a summary: location (INEGI), use, variety, rootstock, planting date and material, number of prunings, Winkler region, yes / no answers to the 67 activities, innovated activities, IS, IR and risk (flagged provisional if the vineyard did not confirm all its knobs), adaptive capacity and its network links without the omnipresent node. Only the summary is stored, in the coordinator's browser.

## 13.2 Frequency of each practice

$$f_a = \Sigma_\ell\,[r_{\ell a} = \text{sí}] \,/\, \Sigma_\ell\,[r_{\ell a} \in \{\text{sí, no}\}] \tag{23}$$

$\ell$ runs over vineyards; unanswered items stay out of the denominator, so a missing answer is not mistaken for "no". The same ratio is computed within each Banxico region, each use and each Winkler region. The atlas shows next to each value how many vineyards answered.

## 13.3 National network

$$W_{ij} = n_{ij}/L\,; \quad V_i = n_i/L \tag{24}$$

$L$: vineyards with a network; $n_{ij}$: vineyards whose network contains link $i$–$j$; $n_i$: vineyards where element $i$ appears. The chapter 3 statistics are computed on this network.

## 13.4 Aggregation test

The app combines two copies of a test vineyard with all 67 activities answered "yes". If aggregation is correct, all 67 practices are 100 % with 2 answers and the national network matches the test vineyard's (72 nodes and 250 links), with all weights equal to 1. The test runs in the Validation block and all three checks pass.

REFERENCE

# A Technical glossary

| | |
|---|---|
| **BBCH**<br>*BBCH* | Decimal scale of phenological stages; for the grapevine, from winter bud (00) to leaf fall (97) (Lorenz et al., 1995). |
| **Brandes algorithm**<br>*Brandes (algoritmo)* | Exact betweenness computation in $O(nm + n^2 \log n)$ for weighted networks (Brandes, 2001). |
| **Adaptive capacity (CA)**<br>*Capacidad adaptativa (CA)* | Mean of regeneration and three Sarandón sub-indicators (diversification, technical knowledge, institutional integration), on 0–1. |
| **Cophenetic correlation**<br>*Correlación cofenética* | Pearson correlation between original distances and the tree's merge heights (Sokal & Rohlf, 1962). |
| **ERA5**<br>*ERA5* | ECMWF global reanalysis (1940 to present, 0.25° cells) reconstructing past climate with a model and the available observations (Hersbach et al., 2020). |
| **Single linkage**<br>*Enlace simple* | Agglomerative clustering in which the distance between groups is that of their closest pair of members (Sneath, 1957). |
| **Exploratory scenario**<br>*Escenario exploratorio* | Possible future built on explicit assumptions, with no assigned probability (Godet & Durance, 2011). |
| **Cool-night index (CI)**<br>*Frescor nocturno (CI)* | Mean September minimum temperature in the northern hemisphere (Tonietto & Carbonneau, 2004). |
| **Winkler degree-days (W)**<br>*Grados día de Winkler (W)* | Σ Apr–Oct $\max(0, T_{mean} - 10) \cdot$ days; five regions (Amerine & Winkler, 1944). |
| **GWP100**<br>*GWP100* | 100-year global warming potential; 273 for $N_2O$ (AR6). |
| **Huglin index (HI)**<br>*Índice de Huglin (HI)* | Σ Apr–Sep of the mean of $(T_{mean} - 10)$ and $(T_{max} - 10)$ times days, times a latitude coefficient (Huglin, 1978). |
| **Incidence**<br>*Incidencia* | Binary elements × activities matrix: 1 if the element takes part in the activity. |

| | |
|---|---|
| **Ratio Viña reading**<br>*Lectura de Ratio Viña* | Value, rule or grouping proposed by the app and not published in the source; it is flagged and editable. |
| **Louvain**<br>*Louvain* | Community detection method maximising modularity through local moves and aggregation (Blondel et al., 2008). |
| **Modularity (Q)**<br>*Modularidad (Q)* | Fraction of weight within communities minus that expected at random (Newman, 2006). |
| **Foresight**<br>*Prospectiva* | Discipline building possible futures to guide present decisions (Berger; de Jouvenel; Godet). |
| **Local knowledge stack**<br>*Stack de conocimiento local* | Layered structure of farming knowledge: rules, steps and activities (Díaz-José et al., 2026). In Ratio Viña, stages take the place of rules. |
| **Structural value**<br>*Valor estructural* | An element's relative participation in activities, used as default value. |
| **Eigenvector centrality**<br>*Vector propio* | Centrality proportional to that of neighbours; obtained by power iteration. |

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Code: github.com/hectortecumshe-ai/RatioVina

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Héctor Tecumshé Mojica-Zárate · Oscar Díaz-José · Edgar Arturo Sánchez-Moreno · Sergio Alejo-Bello · Luis Ángel Barrera-Guzmán

Corresponding author ORCID (Héctor Tecumshé Mojica-Zárate): 0000-0002-9067-3983.

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